

U.S. Airline Performance Analysis

Business Insights and Recommendations Report

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1. Executive Summary

This report covers an analysis of 5.8 million U.S. domestic flight records from 2015, covering 14 airlines and 322 airports. The data was processed using PostgreSQL and SQL, and visualised through a four-page Power BI dashboard covering overall performance, airline comparisons, time-based patterns, and airport-level insights.

The headline finding is that the U.S. airline industry achieved an 82.14% on-time performance rate in 2015, with an average arrival delay of 4.41 minutes across completed flights. While this sits above the 80% industry benchmark, significant variation exists between carriers and across time periods. The overall cancellation rate of 1.54% is within acceptable range, but several individual airlines and airports fall well outside that threshold.

The majority of delays are driven by factors within airline control. Weather, despite being the most publicly cited cause of disruption, accounts for only 54.35% of cancellations but a much smaller share of total delay minutes. Late aircraft propagation and carrier-related issues remain the dominant operational challenges.

2. Key Findings

Overall Performance

Across 5.8 million flights, the industry achieved an 82.14% OTP rate with an average arrival delay of 4.41 minutes. Average departure delay was slightly higher at 9.34 minutes, suggesting airlines recover some time in the air but not consistently enough to offset late departures. Total delay minutes across all completed flights reached 69.1 million.

Airline Performance

Hawaiian Airlines leads the industry with an 89.47% OTP rate and the lowest cancellation rate at 0.22%. Alaska Airlines follows at 87.69%. At the other end, Spirit Air Lines recorded the highest average arrival delay at 14.47 minutes, and American Eagle Airlines had the highest cancellation rate at 5.10%.

Airline	OTP Rate	Avg Arrival Delay	Cancel Rate
Hawaiian Airlines	89.47%	2.02 min	0.22%
Alaska Airlines	87.69%	-0.98 min	0.39%
Delta Air Lines	87.03%	0.19 min	0.44%
Southwest Airlines	81.75%	4.37 min	1.27%
United Air Lines	80.10%	5.43 min	1.27%
JetBlue Airways	78.21%	6.68 min	1.60%
American Eagle Airlines	79.03%	6.46 min	5.10%
Frontier Airlines	74.68%	12.50 min	0.65%
Spirit Air Lines	71.25%	14.47 min	1.71%

Monthly Trends

October recorded the highest OTP at 84% and the lowest average arrival delay at -0.78 minutes, meaning flights were arriving ahead of schedule on average. June was the worst month with a 9.60 minute average arrival delay and an OTP of approximately 73.68%. The pattern reflects the impact of summer travel demand peaking in June and July, with operations recovering through the autumn months.

Day of Week Patterns

Saturday is the best day to fly with an OTP of 84.67% and the lowest cancellation rate. Thursday is the worst performing day for on-time arrivals at 80.80% OTP. Monday has the highest cancellation rate at 2.43%, more than double Friday's rate of 1.02%. This pattern aligns with the surge in business travel at the start of the working week.

Time of Day

The 08:00 to 12:00 window carries the highest volume of flights at 1.46 million departures. The 00:00 to 04:00 slot has the fewest flights at just 0.02 million. Early morning departures in the 04:00 to 08:00 slot tend to have the best on-time performance as delays have not yet accumulated across the network.

Airport Congestion

Atlanta is the busiest airport with 346,836 departures but maintains a reasonable OTP of 84.22%. Chicago O'Hare handles 285,884 departures with a 76.69% OTP and an average departure delay of 14.07 minutes, making it the most operationally stressed major hub. Newark has the highest cancellation rate among major airports at 3.06%.

ILG (Wilmington Airport) recorded the highest average departure delay at 29.57 minutes, but this is partly a reflection of its very low flight volume. Among airports with significant traffic, the congestion pattern is concentrated in Chicago, New York, and Dallas.

Cancellation Analysis

The overall cancellation rate is 1.54%. Weather is the leading cause at 54.35% of all cancellations, followed by airline and carrier issues at 28.11%, National Air System at 17.52%, and Security at 0.02%. While weather dominates cancellations, the airline-driven share at 28% represents a meaningful opportunity for operational improvement since these are within carrier control.

Cancellation Cause	Share of Cancellations
Weather	54.35%
Airline / Carrier	28.11%
National Air System	17.52%
Security	0.02%

3. Root Cause Analysis

The performance gaps in this dataset point to a consistent set of operational causes rather than random variation. For delays, the biggest driver is schedule design. When airlines build tight turnaround times into their schedules, any inbound delay immediately cascades into the next departure. This propagation effect explains why June and July, when flight volumes are highest, also have the worst average delays. There is simply no buffer in the system to absorb disruption when the network is running at full capacity.

Carrier delays, which cover maintenance issues and crew availability, represent the portion of delays that airlines have direct control over. Frontier and Spirit, despite having lower cancellation rates than American Eagle, have among the worst average arrival delays. This suggests their operations struggle with internal execution rather than with external factors like weather or air traffic control.

For cancellations, weather dominates at 54% but this masks an important nuance. Airlines have discretion over when to cancel versus delay a flight. A cancellation decision is often a business and operational choice, not a pure weather response. The fact that airline-driven cancellations account for 28% of the total means that nearly a third of all cancelled flights reflect internal operational failures.

At the airport level, Chicago O'Hare's combination of high volume, tight infrastructure, and severe weather exposure creates a structural delay problem that individual airlines cannot fully solve on their own. However, airlines can reduce their exposure by adjusting scheduling windows at these congested hubs.

4. Business Recommendations

Build Schedule Buffers on High-Delay Routes

Routes and airports with consistently high average delays need scheduled times extended to reflect actual operating conditions. A 10 to 15 minute buffer on chronic delay routes costs almost nothing on the scheduling side but immediately improves reported OTP because the airline stops making promises it cannot reliably keep.

Reduce Carrier-Driven Cancellations

With 28% of cancellations attributed to airline and carrier causes, there is a meaningful opportunity to improve through better maintenance scheduling, proactive aircraft positioning, and crew availability planning. Airlines that invest in predictive maintenance systems and smarter crew rostering tend to show lower carrier cancellation rates over time.

Rebalance Peak Hour Scheduling at Congested Hubs

Chicago O'Hare, Dallas/Fort Worth, and New York airports carry the highest congestion loads. Moving a portion of departures from peak morning and evening windows to off-peak slots reduces competition for runway and gate capacity. Even modest reductions in peak concentration can improve taxi-out times and reduce downstream propagation delays.

Target June and July Operationally

June records the worst monthly delay performance at 9.60 minutes average arrival delay. Summer operations need dedicated planning, including additional ground staff, extended maintenance windows during turnarounds, and contingency crew positioning at high-volume hubs. Treating June and July as high-risk operational periods rather than standard months is a straightforward and low-cost improvement.

Address Monday Cancellation Concentration

Monday has a cancellation rate of 2.43%, more than double Friday. This is a predictable, recurring pattern that should drive staffing and aircraft positioning decisions every week. Positioning spare aircraft and additional crew at major hubs on Sunday evenings specifically to cover Monday morning operations would reduce the cancellation rate on the highest-risk day.

5. Expected Impact

These recommendations address specific, measurable problems identified in the data. The table below summarises expected improvements and realistic timelines.

Recommendation	Expected Improvement	Timeline
Schedule buffers on delay routes	+2 to 3 OTP points	6 months
Carrier cancellation reduction	Reduce carrier share from 28% toward 20%	12 months
Peak hour rebalancing at ORD, DFW, EWR	+1 to 2 OTP points	18 months
Summer operations planning	Reduce June avg delay from 9.6 to under 7 min	Next season
Monday crew and aircraft positioning	Reduce Monday cancel rate from 2.43% toward 1.5%	3 months

If the first three recommendations are implemented together, a combined OTP improvement of 3 to 5 percentage points is achievable within 18 months. The industry is already operating above the 80% benchmark at 82.14%, but the variation between carriers shows there is significant room to raise the floor. Hawaiian Airlines at 89.47% demonstrates what is operationally possible. Closing even half the gap between the worst and best performers would represent a meaningful improvement in passenger experience and operational cost efficiency.

The most important takeaway from this analysis is that most of the performance problems visible in the data are operational and therefore fixable. Weather is real, but it explains a smaller share of disruptions than airlines typically communicate publicly. The bigger opportunities are in scheduling design, carrier execution, and targeted planning for predictable high-risk periods.

Dataset: 5,819,079 U.S. domestic flight records, 2015. Source: U.S. Department of Transportation. Tools: PostgreSQL, SQL, Power BI, DAX. Prepared by Raju Kumar S, Data Analyst, Bengaluru, India.